

Life history, immunity, Peto's paradox and tumours in birds.

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Cancer and tumours may evolve in response to life-history trade-offs between growth and duration of development on one hand, and between growth and maintenance of immune function on the other. Here, we tested whether (i) bird species with slow developmental rates for their body size experience low incidence of tumours because slow development allows for detection of rapid proliferation of cell lineages. We also test whether (ii) species with stronger immune response during development are more efficient at detecting tumour cells and hence suffer lower incidence of tumours. Finally, we tested Peto's paradox, that there is a positive relationship between tumour incidence and body mass. We used information on developmental rates and body mass from the literature and of tumour incidence (8468 birds) and size of the bursa of Fabricius for 7659 birds brought to a taxidermist in Denmark. We found evidence of the expected negative relationship between incidence of tumours and developmental rates and immunity after controlling for the positive association between tumour incidence and body size. These results suggest that evolution has modified the incidence of tumours in response to life history and that Peto's paradox may be explained by covariation between body mass, developmental rates and immunity.

The spontaneous immortalization probability of mammalian cell culture strains, as their proliferative capacity, correlates with species body mass, not longevity.

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The Peto's paradox consists in the observation that individuals from long-lived and large animal species do not experience a higher cancer incidence, despite being exposed for longer time to the possibility of accumulating mutations and having more target cells exposed to the phenomenon. The existence of this paradox has been recently confirmed (Vincze et al., 2022). Concurrently, robust evidence has been published that longevity involves a convergent evolution of cellular mechanisms that prevent the accumulation of mutations (Cagan et al., 2022). It remains unclear which cellular mechanisms are critical to allow the evolution of a large body mass while keeping cancer at bay. Adding to existing data linking cellular replicative potential and species body mass (Lorenzini et al., 2005), we have grown a total of 84 skin fibroblast cell strains from 40 donors of 17 mammalian species and analyzed their Hayflick's limit, i.e., their senescent plateau, and eventual spontaneous immortalization escape. The correlation of immortalization and replicative capacity of the species with their longevity, body mass and metabolism has been assessed through phylogenetic multiple linear regression (MLR). The immortalization probability is negatively related to species body mass. The new evaluation and additional data about replicative potential strengthen our previous observation, confirming that stable and extended proliferation is strongly correlated with the evolution of a large body mass rather than lifespan. The relation between immortalization and body mass suggests a need to evolve stringent mechanisms that control genetic stability during the evolution of a large body mass.

From Churchill to Elephants: The Role of Protective Genes against Cancer.

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Richard Peto's paradox, first described in 1975 from an epidemiological perspective, established an inverse correlation between the probability of developing cancer in multicellular organisms and the number of cells. Larger animals exhibit fewer tumors compared to smaller ones, though exceptions exist. Mice are more susceptible to cancer than humans, while elephants and whales demonstrate significantly lower cancer prevalence rates than humans. How nature and evolution have addressed the issue of cancer in the animal kingdom remains largely unexplored. In the field of medicine, much attention has been devoted to cancer-predisposing genes, as they offer avenues for intervention, including blocking, downregulating, early diagnosis, and targeted treatment. Predisposing genes also tend to manifest clinically earlier and more aggressively, making them easier to identify. However, despite significant strides in modern medicine, the role of protective genes lags behind. Identifying genes with a mild predisposing effect poses a significant challenge. Consequently, comprehending the protective function conferred by genes becomes even more elusive, and their very existence is subject to questioning. While the role of variable expressivity and penetrance defects of the same variant in a family is well-documented for many hereditary cancer syndromes, attempts to delineate the function of protective/modifier alleles have been restricted to a few instances. In this review, we endeavor to elucidate the role of protective genes observed in the animal kingdom, within certain genetic syndromes that appear to act as cancer-resistant/repressor alleles. Additionally, we explore the role of protective alleles in conditions predisposing to cancer. The ultimate goal is to discern why individuals, like Winston Churchill, managed to live up to 91 years of age, despite engaging in minimal physical activity, consuming large quantities of alcohol daily, and not abstaining from smoking.

Importance of metabolic rate to the relationship between the number of genes in a functional category and body size in Peto's paradox for cancer.

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Elucidation of tumour suppression mechanisms is a major challenge in cancer biology. Therefore, Peto's paradox, or low cancer incidence in large animals, has attracted focus. According to the gene-abundance hypothesis, which considers the increase/decrease in cancer-related genes with body size, researchers evaluated the associations between gene abundance and body size. However, previous studies only focused on a few specific gene functions and have ignored the alternative hypothesis (metabolic rate hypothesis): in this hypothesis, the cellular metabolic rate and subsequent oxidative stress decreases with increasing body size. In this study, we have elected to explore the gene-abundance hypothesis taking into account the metabolic rate hypothesis. Thus, we comprehensively investigated the correlation between the number of genes in various functional categories and body size while at the same time correcting for the mass-specific metabolic rate (B_c). A number of gene functions that correlated with body size were initially identified, but they were found to be artefactual due to the decrease in B_c with increasing body size. By contrast, immune system-related genes were found to increase with increasing body size when the correlation included this correction for B_c . These findings support the gene-abundance hypothesis and emphasize the importance of also taking into account the metabolic rate when evaluating gene abundance-body size relationships. This finding may be useful for understanding cancer evolution and tumour suppression mechanisms as well as for determining cancer-related genes and functions.

Oxford and the Savannah: can the hippo provide an explanation for Peto's paradox?

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Peto's paradox is the counterintuitive finding that increasing body mass and thereby cell number does not correlate with an increase in cancer incidence across different species. The Hippo signaling pathway is an evolutionarily conserved system that determines organ size by regulating apoptosis and cell proliferation. It also affects cell growth by microRNA-29 (miR-29)-mediated cross-talk to the mTOR signaling pathway. Whether these pathways that decide organ size could explain this paradox merits consideration. Inactivation of most genes of the Hippo pathway in *Drosophila melanogaster* genetic screens causes excessive tissue-specific growth of developing tissues. Altered Hippo pathway activity is frequently found in diverse tumor types, but mutations of component pathway genes are rare. Most Hippo pathway components are encoded by tumor suppressor genes (TSG), but an exception is the downstream effector gene called YAP. Activity of the Hippo pathway causes deactivating phosphorylation of YES-associated protein (YAP) with nuclear exclusion. YAP can also be phosphorylated at a second site, S127, by AKT. YAP induces the expression of genes responsible for proliferation and suppression of apoptosis. Resolving Peto's paradox may serendipitously provide new insights into the biology and treatment of cancer. This article considers Hippo signaling and Peto's paradox in the context of TSG-oncogene computed models. Interspecies differences in dietary composition, metabolic rates, and anabolic processes are also discussed in the context of Hippo-mTOR signaling. The metabolically important LKB1-AMPK (liver kinase B1-AMP activated protein kinase) signaling axis that suppresses the mTOR pathway is also considered.

A mathematical solution to Peto's paradox using Polya's urn model: implications for the aetiology of cancer in general.

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Ageing is the leading risk factor for the emergence of cancer in humans. Accumulation of pro-carcinogenic events throughout life is believed to explain this observation; however, the lack of direct correlation between the number of cells in an organism and cancer incidence, known as Peto's Paradox, is at odds with this assumption. Finding the events responsible for this discrepancy can unveil mechanisms with potential uses in prevention and treatment of cancer in humans. On the other hand, the immune system is important in preventing the development of clinically relevant tumours by maintaining a fine equilibrium between reactive and suppressive lymphocyte clones. It is suggested here that the loss of this equilibrium is what ultimately leads to increased risk of cancer and to propose a mechanism for the changes in clonal proportions based on decreased proliferative capacity of lymphocyte clones as a natural phenomenon of ageing. This mechanism, being a function of the number of cells, provides an explanation for Peto's Paradox.

Novel Genomic Insights into Body Size Evolution in Cetaceans and a Resolution of Peto's Paradox.

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AbstractCetaceans (whales, dolphins, and porpoises) have undergone a radical transformation from the typical terrestrial mammalian body plan to a streamlined body, while exhibiting dramatic interspecific size differences. However, the molecular mechanisms underlying the diversification of cetacean body size

are largely unknown. Here, by using genomic and phenotypic data for 22 cetaceans, we performed phylogenetic genome-body size analysis and explored the genetic basis of the high diversity of body size in cetaceans. A functional enrichment analysis showed that body size-related genes in cetaceans are enriched in pathways associated with immunity, cell growth, and metabolism, suggesting that they contributed to body size diversification. Genes showing correlated evolution with body size were mainly involved in immune surveillance, tumor suppression function, and development of hypertumors. The role of these genes in tumor control resolves Peto's paradox (i.e., the lack of a correspondence between an expansion in body size and, thereby, cell number and an increased cancer incidence). Our results provide novel insights into the evolution of substantial body size variation in cetaceans.

TP53 copy number expansion is associated with the evolution of increased body size and an enhanced DNA damage response in elephants.

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A major constraint on the evolution of large body sizes in animals is an increased risk of developing cancer. There is no correlation, however, between body size and cancer risk. This lack of correlation is often referred to as 'Peto's Paradox'. Here, we show that the elephant genome encodes 20 copies of the tumor suppressor gene TP53 and that the increase in TP53 copy number occurred coincident with the evolution of large body sizes, the evolution of extreme sensitivity to genotoxic stress, and a hyperactive TP53 signaling pathway in the elephant (Proboscidean) lineage. Furthermore, we show that several of the TP53 retrogenes (TP53RTGs) are transcribed and likely translated. While TP53RTGs do not appear to directly function as transcription factors, they do contribute to the enhanced sensitivity of elephant cells to DNA damage and the induction of apoptosis by regulating activity of the TP53 signaling pathway. These results suggest that an increase in the copy number of TP53 may have played a direct role in the evolution of very large body sizes and the resolution of Peto's paradox in Proboscideans.

Size, longevity and cancer: age structure.

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There is significant recent interest in Peto's paradox and the related problem of the evolution of large, long-lived organisms in terms of cancer robustness. Peto's paradox refers to the expectation that large, long-lived organisms have a higher lifetime cancer risk, which is not the case: a paradox. This paradox, however, is circular: large, long-lived organisms are large and long-lived because they are cancer robust. Lifetime risk, meanwhile, depends on the age distributions of both cancer and competing risks: if cancer strikes before competing risks, then lifetime risk is high; if not, not. Because no set of competing risks is generally prevalent, it is instructive to temporarily dispose of competing risks and investigate the pure age dynamics of cancer under the multistage model of carcinogenesis. In addition to augmenting earlier results, I show that in terms of cancer-free lifespan large organisms reap greater benefits from an increase in cellular cancer robustness than smaller organisms. Conversely, a higher cellular cancer robustness renders cancer-free lifespan more resilient to an increase in size. This interaction may be an important driver of the evolution of large, cancer-robust organisms.

A phylogenetic review of cancer resistance highlights evolutionary solutions to Peto's Paradox.

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Cancer is a genetic disease present in all complex multicellular lineages. Finding ways to eliminate it is a goal of a large part of the scientific community and nature itself. Early, scientists realized that the cancer incidence at the species level was not related to the number of cells or lifespan, a phenomenon called Peto's Paradox. The interest in resolving this paradox triggered a growing interest in investigating the natural strategies for cancer suppression hidden in the animal's genomes. Here, we gathered information on the main mechanisms that confer resistance to cancer, currently described for lineages that have representatives with extended longevity and large body sizes. Some mechanisms to reduce or evade cancer are common and shared between lineages, while others are species-specific. The diversity of paths that evolution followed to face the cancer challenge involving coding, regulatory, and structural aspects of genomes is astonishing and much yet lacks discovery. Multidisciplinary studies involving oncology, ecology, and evolutionary biology and focusing on nonmodel species can greatly expand the frontiers of knowledge about cancer resistance in animals and may guide new promising treatments and prevention that might apply to humans.
